

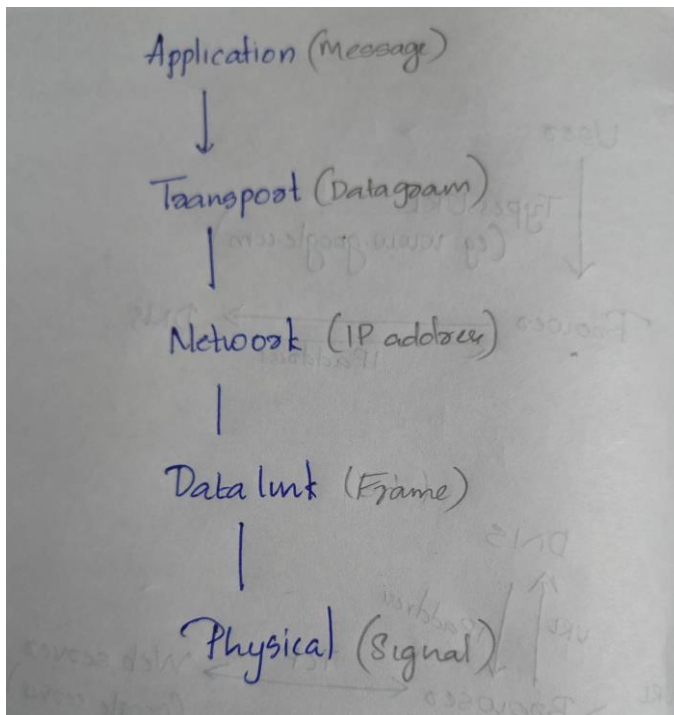
Assignment 1

PCCST501 – Computer Networks

How the Internet Delivers a Web Page

Choose one website that you frequently use (e.g., Google, Amazon, YouTube, Gmail, Instagram, or your college website). Explain the complete process that occurs when a user types the website URL into a web browser and presses Enter. Answer the following questions.

1. Draw the TCP/IP protocol stack and explain the role of each layer.



Application Layer

The logical connection between the two application layers is end-to-end. The two application layers exchange messages between each other as though there were a bridge between the two layers. Communication at the application layer is between two processes (two programs running at this layer). To communicate, a process sends a request to the other process and receives a response. Process-to-process communication is the duty of the application layer.

Transport Layer

The logical connection at the transport layer is also end-to-end. The transport layer at the source host gets the message from the application layer, encapsulates it in a transport layer packet (called a segment or a user datagram in different protocols) and sends it, through the logical (imaginary) connection, to the transport layer at the destination host.

Network Layer

The network layer is responsible for creating a connection between the source computer and the destination computer. The communication at the network layer is host-to-host. However, since there can be several routers from the source to the destination, the routers in the path are responsible for choosing the best route for each packet. The network layer in the Internet includes the main protocol, Internet Protocol (IP), that defines the format of the packet, called a datagram at the network layer. IP also defines the format and the structure of addresses used in this layer.

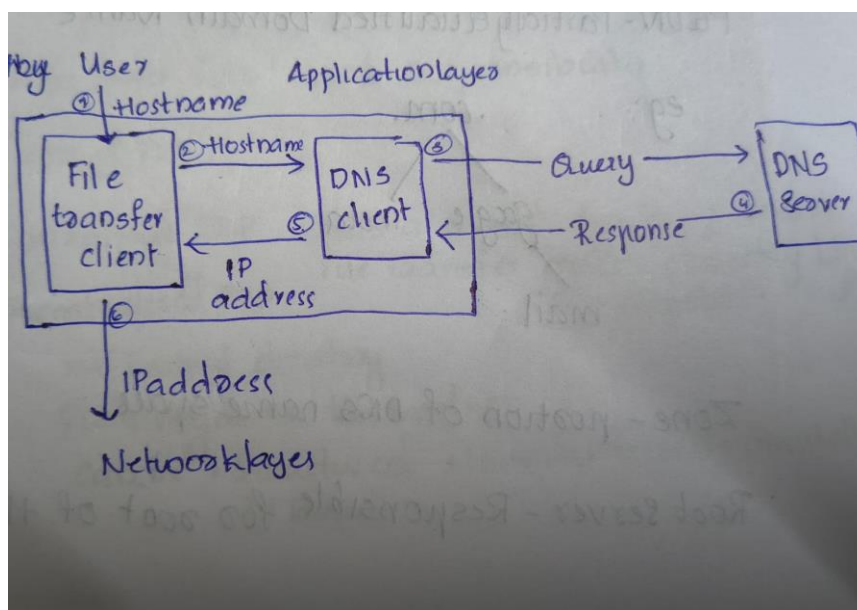
Datalink Layer

There may be several overlapping sets of links that a datagram can travel from the host to the destination. The routers are responsible for choosing the best links. However, when the next link to travel is determined by the router, the data-link layer is responsible for taking the datagram and moving it across the link.

Physical Layer

the physical layer is the lowest level in the TCP/IP protocol suite. Two devices are connected by a transmission medium (cable or air). We need to know that the transmission medium does not carry bits; it carries electrical or optical signals. So the bits received in a frame from the data-link layer are transformed and sent through the transmission media.

2. Explain the role of DNS in converting the domain name into an IP address.



The following six steps map the host name to an IP address:

1. The user passes the host name to the file transfer client.
2. The file transfer client passes the host name to the DNS client.
3. Each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.
5. The DNS client passes the IP address to the file transfer server.
6. The file transfer client now uses the received IP address to access the file transfer server.

3. Describe how HTTP is used to request and receive the webpage.

The HyperText Transfer Protocol (HTTP) is a protocol that is used to define how the client-server programs can be written to retrieve web pages from the Web. An HTTP client sends a request; an HTTP server returns a response. The server uses the port number 80; the client uses a temporary port number. HTTP uses the services of TCP.

An HTTP request message contains:

- 1 Request line
- 2 Zero or more header lines
- 3 Blank line
- 4 Optional body

An HTTP response message contains:

- 1 Status line
- 2 Zero or more header lines
- 3 Blank line
- 4 Optional body

Working of HTTP:

1. The user enters URL `www.google.com` in the web browser.
2. The browser (HTTP client) establishes a TCP connection with the Google web server.
3. The client sends an **HTTP request** for the required webpage.
4. The server processes the request and sends an **HTTP response** containing the requested webpage (or an error message if the page is not found).

5. The browser receives the response and displays the webpage to the user.

4. Explain how TCP provides reliable communication for HTTP.

The HyperText Transfer Protocol (HTTP) defines how client-server programs communicate to retrieve web pages from the Web. An HTTP client sends a request. An HTTP server returns a response. HTTP uses the services of TCP. TCP is: Connection-oriented and Reliable.

Before an HTTP transaction: A TCP connection must be established.

After the transaction: The connection may be terminated.

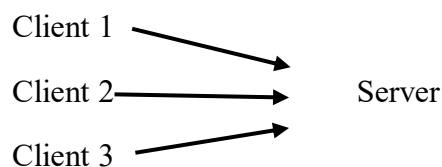
TCP provides :

- Reliable delivery by ensuring that all data reaches the destination correctly.
- Error control, which detects lost or damaged segments and retransmits them.
- Flow control, which prevents the sender from transmitting data faster than the receiver can process it.
- Congestion control, which reduces network congestion by controlling the transmission rate.

5. Identify whether the application follows a Client–Server or Peer-to-Peer architecture and justify your answer.

Applications like HTTP, FTP, SSH, and E-mail are examples of the Client–Server paradigm: A client sends a request for a service, while a **server** waits continuously to receive requests and provide the requested service. The server is always running and can serve multiple clients simultaneously.

In HTTP, the web browser acts as the client and the web server provides web pages. Here the Google web server receives the request from the browser and provide corresponding response that is the requested web page.



6. Compare HTTP and FTP with at least four differences.

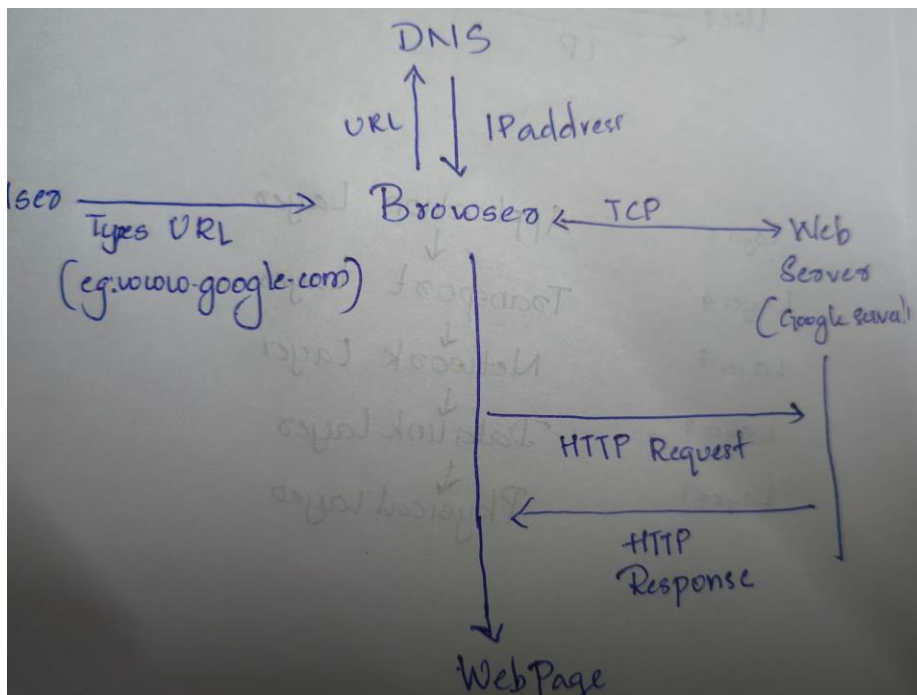
Feature	HTTP (HyperText Transfer Protocol)	FTP (File Transfer Protocol)
Purpose	Used to transfer web pages and web content between a browser and a web server.	Used to transfer files between computers over a network.
Connections	Uses a single TCP connection for both commands and data.	Uses two TCP connections: one control connection and one data connection.
Port Numbers	Uses TCP Port 80	Uses TCP Port 21 for control and TCP Port 20 for data.
Operations	Mainly retrieves web resources	Supports file upload, download, rename, delete, and directory management.
Usage	Web browsing and accessing websites.	Uploading, downloading, and managing files on remote servers

7. Explain how cookies improve user experience with one real-world example.

When a client sends a request to a server, the browser looks in the cookie directory to see if it can find a cookie sent by that server. If found, the cookie is included in the request. When the server receives the request, it knows that this is an old client, not a new one. The contents of the cookie are never read by the browser or disclosed to the user. It is a cookie made by the server and eaten by the server.

Example: An electronic store (e-commerce) can use a cookie for its client shoppers. When a client selects an item and inserts it in a cart, a cookie that contains information about the item, such as its number and unit price, is sent to the browser. If the client selects a second item, the cookie is updated with the new selection information, and so on. When the client finishes shopping and wants to check out, the last cookie is retrieved and the total charge is calculated.

8. Draw a flow diagram showing the complete communication from the browser to the web server and back.



The user types www.google.com in the browser. The browser checks whether the webpage is already stored in its cache. If a cached copy exists and is still valid, it loads immediately. Otherwise, the browser starts communicating over the Internet. The browser needs the IP address of www.google.com. The DNS server returns the IP address of Google's web server. Using the obtained IP address, the browser establishes a TCP connection with the web server. The browser sends an HTTP request. The web server processes the request and returns the requested web page.

9. Write a brief conclusion (100–150 words) describing the importance of application layer protocols in everyday Internet usage.

The application layer is different from other layers in that it is the highest layer in the suite. The protocols in this layer do not provide services to any other protocol in the suite; they only receive services from the protocols in the transport layer. This means that protocols can be removed from this layer easily. New protocols can be also added to this layer as long as the new protocol can use the service provided by one of the transport-layer protocols. Since the application layer is the only layer that provides services to the Internet user, the flexibility of the application layer allows new application protocols to be easily added to the Internet.

Application-layer protocols are important because they allow users to access and use Internet services easily. They provide a set of rules that helps different applications communicate with each other over a network. For example, HTTP is used for browsing websites, FTP is used to transfer files, SMTP, POP, and IMAP are used for sending and receiving emails, and DNS converts website names into IP addresses so that computers can find the correct servers.

These protocols support both Client–Server and Peer-to-Peer (P2P) communication models, making it possible to use different types of Internet applications. They also ensure that devices with different operating systems and hardware can communicate without any compatibility issues.

Without application-layer protocols, everyday Internet activities such as browsing the web, sending emails, downloading files, and accessing online services would not be possible. Therefore, application-layer protocols are an essential part of computer networks and play a major role in making Internet communication simple, reliable, and efficient.